

AthlonX Gym Management System - Database Structure Report

Generated: January 8, 2026
Database: Oracle 23ai FREE
Schema: SYSTEM
Total Application Tables: 25

Database Overview

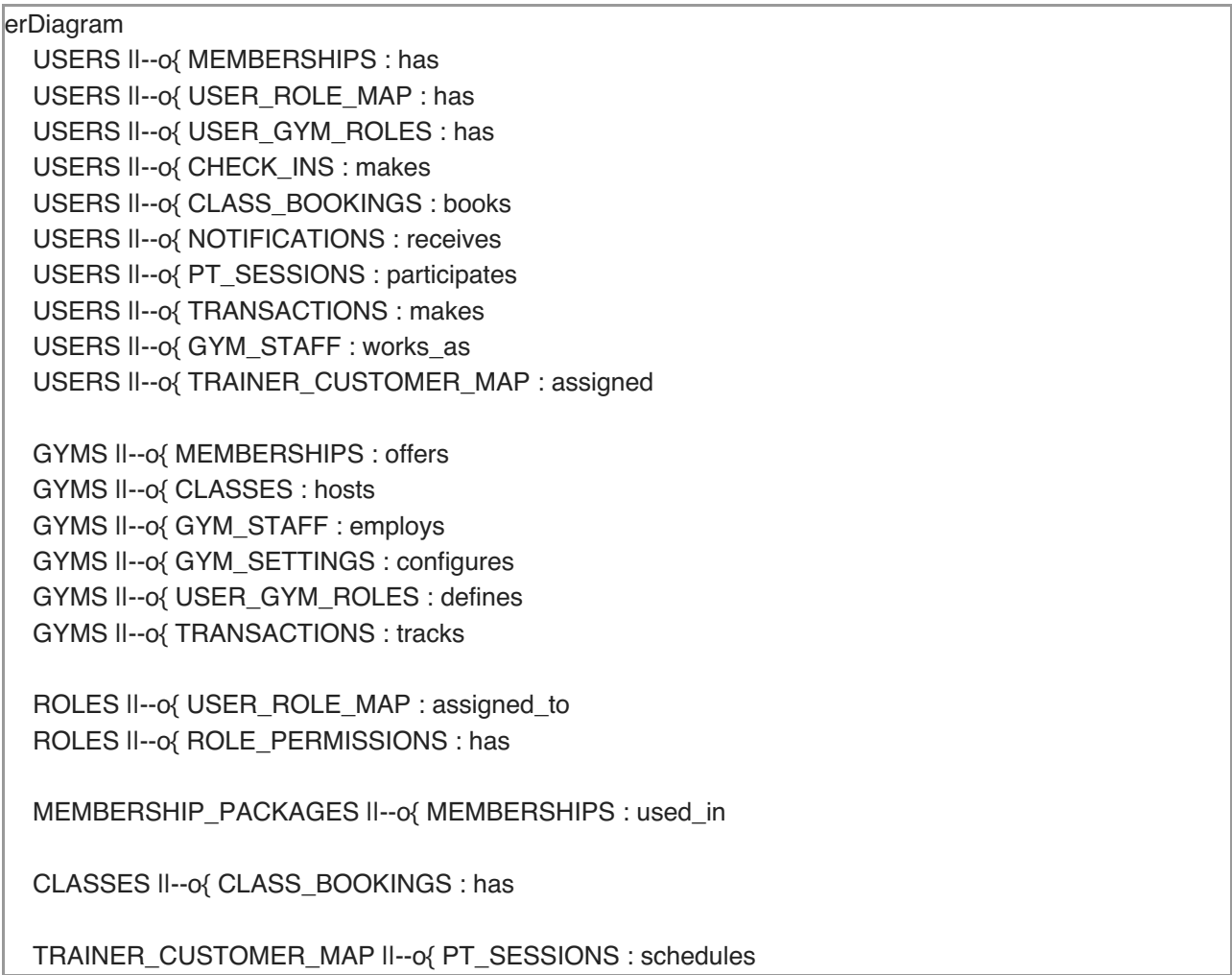


Table Summary

Table	Records	Description
USERS	170	All system users (members, trainers, staff, admins)
GYMS	1	Gym locations/branches
MEMBERSHIPS	103	Active/inactive member subscriptions
MEMBERSHIP_PACKAGES	5	Available membership plans

CLASSES	-	Group fitness classes
CLASS_BOOKINGS	-	Class reservations
CHECK_INS	-	Member gym entry logs
ROLES	4	System roles (ADMIN, TRAINER, MEMBER, etc.)
USER_ROLE_MAP	158	User-to-role assignments
USER_GYM_ROLES	-	Gym-specific role assignments
TRAINER_CUSTOMER_MAP	19	Trainer-member assignments
PT_SESSIONS	6	Personal training sessions
TRANSACTIONS	51	Payment records
STAFF_SHIFTS	4	Staff work schedules
STAFF_PERFORMANCE	4	Staff KPI tracking
GYM_SETTINGS	10	Configuration settings
NOTIFICATIONS	-	User notifications
ALERTS	-	System alerts
ROLE_APPLICATIONS	-	Role upgrade requests
ROLE_CHANGE_AUDIT	-	Role change history
ROLE_PERMISSIONS	-	Permission assignments
OTP_CODES	-	One-time passwords
BLACKOUT_DAYS	3	Gym closure dates
PROGRESS_NOTES	-	Member progress tracking

Detailed Table Structures

USERS (Core User Table)

Column	Type	Length	Nullable	Description
USER_ID	NUMBER	22	NO	Primary Key
USERNAME	VARCHAR2	100	NO	Unique login name
PASSWORD	VARCHAR2	255	NO	Encrypted password
FULL_NAME	VARCHAR2	100	YES	Display name
EMAIL	VARCHAR2	100	YES	Email address
PHONE	VARCHAR2	1020	YES	Phone number
PHONE_NUMBER	VARCHAR2	20	YES	Alternate phone
STATUS	VARCHAR2	1020	YES	Account status
AVATAR_ID	VARCHAR2	1020	YES	Profile picture
GOOGLE_ID	VARCHAR2	255	YES	OAuth Google ID
FACEBOOK_ID	VARCHAR2	255	YES	OAuth Facebook ID
AUTH_PROVIDER	VARCHAR2	20	YES	Authentication method
IS_FIRST_LOGIN	NUMBER	22	YES	First login flag
ACCOUNT_NON_LOCKED	NUMBER	22	YES	Lock status
CREATED_AT	TIMESTAMP	11	YES	Registration date
CREATED_BY	NUMBER	22	YES	Creator user ID
LEAVING_DATE	DATE	7	YES	Departure date
PASSWORD_CHANGED_AT	TIMESTAMP	11	YES	Last password change

Indexes: UK_USERS_GOOGLE_ID, UK_USERS_FACEBOOK_ID, UK_USERS_PHONE, IDX_USERS_EMAIL

GYMS

Column	Type	Length	Nullable	Description
GYM_ID	NUMBER	22	NO	Primary Key
GYM_NAME	VARCHAR2	100	NO	Gym name
ADDRESS	VARCHAR2	255	YES	Physical address
PHONE	VARCHAR2	20	YES	Contact phone
EMAIL	VARCHAR2	100	YES	Contact email
OWNER_ID	NUMBER	22	NO	FK → USERS
CREATED_AT	TIMESTAMP	11	YES	Creation date

MEMBERSHIPS

Column	Type	Length	Nullable	Description
MEMBERSHIP_ID	NUMBER	22	NO	Primary Key
USER_ID	NUMBER	22	NO	FK → USERS
GYM_ID	NUMBER	22	YES	FK → GYMS
PACKAGE_ID	NUMBER	22	NO	FK → MEMBERSHIP_PACKAGES
START_DATE	DATE	7	NO	Subscription start
END_DATE	DATE	7	NO	Subscription end
STATUS	VARCHAR2	20	YES	ACTIVE/EXPIRED/CANCELLED
CREATED_AT	TIMESTAMP	11	YES	Record creation

Indexes: IDX_MEMBERSHIPS_END_DATE

MEMBERSHIP_PACKAGES

Column	Type	Length	Nullable	Description
PACKAGE_ID	NUMBER	22	NO	Primary Key
PACKAGE_NAME	VARCHAR2	100	NO	Plan name
DESCRIPTION	VARCHAR2	500	YES	Plan details
PRICE	NUMBER	22	NO	Monthly price
DURATION_MONTHS	NUMBER	22	NO	Duration
IS_ACTIVE	NUMBER	22	YES	Availability flag

Indexes: IDX_MP_ACTIVE

CLASSES

Column	Type	Length	Nullable	Description
CLASS_ID	NUMBER	22	NO	Primary Key
CLASS_NAME	VARCHAR2	100	NO	Class title

DESCRIPTION	VARCHAR2	500	YES	Class details
TRAINER_ID	NUMBER	22	YES	FK → USERS (trainer)
GYM_ID	NUMBER	22	YES	FK → GYMS
SCHEDULE_TIME	TIMESTAMP	11	YES	Class time
DURATION_MINUTES	NUMBER	22	YES	Duration
MAX_CAPACITY	NUMBER	22	YES	Seat limit
CURRENT_BOOKINGS	NUMBER	22	YES	Current count

CLASS_BOOKINGS

Column	Type	Length	Nullable	Description
BOOKING_ID	NUMBER	22	NO	Primary Key
USER_ID	NUMBER	22	NO	FK → USERS
CLASS_ID	NUMBER	22	NO	FK → CLASSES
GYM_ID	NUMBER	22	YES	FK → GYMS
STATUS	VARCHAR2	20	YES	CONFIRMED/CANCELLED
BOOKED_AT	TIMESTAMP	11	YES	Booking time

CHECK_INS

Column	Type	Length	Nullable	Description
CHECK_IN_ID	NUMBER	22	NO	Primary Key
USER_ID	NUMBER	22	NO	FK → USERS
CHECK_IN_TIME	TIMESTAMP	11	NO	Entry time
CHECK_OUT_TIME	TIMESTAMP	11	YES	Exit time

ROLES

Column	Type	Length	Nullable	Description
ROLE_ID	NUMBER	22	NO	Primary Key
ROLE_NAME	VARCHAR2	50	NO	ADMIN, TRAINER, MEMBER, STAFF

USER_ROLE_MAP

Column	Type	Length	Nullable	Description
MAP_ID	NUMBER	22	NO	Primary Key
USER_ID	NUMBER	22	NO	FK → USERS
ROLE_ID	NUMBER	22	NO	FK → ROLES

USER_GYM_ROLES (Multi-Gym Role Support)

Column	Type	Length	Nullable	Description
ID	NUMBER	22	NO	Primary Key
USER_ID	NUMBER	22	NO	FK → USERS

GYM_ID	NUMBER	22	NO	FK → GYMS
ROLE	VARCHAR2	20	NO	Role at this gym
STATUS	VARCHAR2	20	NO	ACTIVE/INACTIVE
GRANTED_BY	NUMBER	22	YES	FK → USERS
GRANTED_AT	TIMESTAMP	11	YES	Grant date
EXPIRES_AT	TIMESTAMP	11	YES	Expiration
NOTES	VARCHAR2	500	YES	Comments

Indexes: IDX_UGR_USER, IDX_UGR_GYM, IDX_UGR_ROLE, IDX_UGR_STATUS, UK_USER_GYM_ROLE

TRAINER_CUSTOMER_MAP

Column	Type	Length	Nullable	Description
MAP_ID	NUMBER	22	NO	Primary Key
TRAINER_USER_ID	NUMBER	22	NO	FK → USERS (trainer)
CUSTOMER_USER_ID	NUMBER	22	NO	FK → USERS (member)

PT_SESSIONS (Personal Training)

Column	Type	Length	Nullable	Description
SESSION_ID	NUMBER	22	NO	Primary Key
TRAINER_ID	NUMBER	22	NO	FK → USERS
MEMBER_ID	NUMBER	22	NO	FK → USERS
GYM_ID	NUMBER	22	YES	FK → GYMS
SESSION_DATE	TIMESTAMP	11	NO	Schedule time
DURATION_MINUTES	NUMBER	22	NO	Session length
STATUS	VARCHAR2	20	NO	SCHEDULED/COMPLETED/CANCELLED
PROGRESS_NOTES	CLOB	4000	YES	Session notes
WORKOUT_PLAN	CLOB	4000	YES	Exercise plan
DIET_PLAN	CLOB	4000	YES	Nutrition plan
IS_RECURRING	NUMBER	22	YES	Recurring flag
RECURRING_FREQUENCY	VARCHAR2	20	YES	WEEKLY/BIWEEKLY
CREATED_AT	TIMESTAMP	11	YES	Creation date
UPDATED_AT	TIMESTAMP	11	YES	Last update

Indexes: IDX_PTS_TRAINER, IDX_PTS_MEMBER, IDX_PTS_DATE, IDX_PTS_STATUS

TRANSACTIONS

Column	Type	Length	Nullable	Description
TRANSACTION_ID	NUMBER	22	NO	Primary Key
USER_ID	NUMBER	22	YES	FK → USERS
GYM_ID	NUMBER	22	YES	FK → GYMS
AMOUNT	NUMBER	22	YES	Transaction amount
CATEGORY	VARCHAR2	1020	YES	MEMBERSHIP/PT/CLASS

STATUS	VARCHAR2	1020	YES	SUCCESS/PENDING/FAILED
DESCRIPTION	VARCHAR2	1020	YES	Details
DATE_TIME	TIMESTAMP	11	NO	Transaction time

GYM_STAFF

Column	Type	Length	Nullable	Description
STAFF_ID	NUMBER	22	NO	Primary Key
USER_ID	NUMBER	22	NO	FK → USERS
GYM_ID	NUMBER	22	NO	FK → GYMS
POSITION	VARCHAR2	50	YES	Job title
HIRE_DATE	DATE	7	YES	Start date
SALARY	NUMBER	22	YES	Compensation
STATUS	VARCHAR2	20	YES	ACTIVE/INACTIVE

STAFF_PERFORMANCE

Column	Type	Length	Nullable	Description
PERFORMANCE_ID	NUMBER	22	NO	Primary Key
STAFF_ID	NUMBER	22	NO	FK → USERS
RECORD_MONTH	VARCHAR2	7	NO	YYYY-MM
SESSIONS_COMPLETED	NUMBER	22	YES	Sessions count
ATTENDANCE_RATE	NUMBER	22	YES	Attendance %
SATISFACTION_SCORE	NUMBER	22	YES	Customer rating
PRESENT_DAYS	NUMBER	22	YES	Days present
ABSENT_DAYS	NUMBER	22	YES	Days absent
LATE_DAYS	NUMBER	22	YES	Days late

Unique Constraint: UK_STAFF_MONTH (STAFF_ID + RECORD_MONTH)

STAFF_SHIFTS

Column	Type	Length	Nullable	Description
SHIFT_ID	NUMBER	22	NO	Primary Key
STAFF_ID	NUMBER	22	NO	FK → USERS
SHIFT_DATE	DATE	7	NO	Work date
START_TIME	TIMESTAMP	11	NO	Shift start
END_TIME	TIMESTAMP	11	NO	Shift end
STATUS	VARCHAR2	20	YES	SCHEDULED/COMPLETED

GYM_SETTINGS

Column	Type	Length	Nullable	Description
SETTING_ID	NUMBER	22	NO	Primary Key
SETTING_TYPE	VARCHAR2	50	NO	Setting category

SETTING_KEY	VARCHAR2	100	NO	Setting name
SETTING_VALUE	CLOB	4000	YES	Value (JSON/text)

NOTIFICATIONS

Column	Type	Length	Nullable	Description
NOTIFICATION_ID	NUMBER	22	NO	Primary Key
USER_ID	NUMBER	22	NO	FK → USERS
TITLE	VARCHAR2	200	NO	Notification title
MESSAGE	CLOB	4000	YES	Content
IS_READ	NUMBER	22	YES	Read status
CREATED_AT	TIMESTAMP	11	YES	Send time

ROLE_APPLICATIONS

Column	Type	Length	Nullable	Description
ID	NUMBER	22	NO	Primary Key
USER_ID	NUMBER	22	NO	FK → USERS (applicant)
GYM_ID	NUMBER	22	NO	FK → GYMS
REQUESTED_ROLE	VARCHAR2	20	NO	Target role
STATUS	VARCHAR2	20	NO	PENDING/APPROVED/REJECTED
CREDENTIALS	CLOB	4000	YES	Supporting docs
REVIEWED_BY	NUMBER	22	YES	FK → USERS (reviewer)
REVIEWED_AT	TIMESTAMP	11	YES	Review date
REJECTION_REASON	VARCHAR2	500	YES	Decline reason
APPLIED_AT	TIMESTAMP	11	YES	Application date

ROLE_CHANGE_AUDIT

Column	Type	Length	Nullable	Description
ID	NUMBER	22	NO	Primary Key
USER_ID	NUMBER	22	NO	Affected user
GYM_ID	NUMBER	22	NO	FK → GYMS
ACTION	VARCHAR2	30	NO	GRANT/REVOKE/MODIFY
OLD_ROLE	VARCHAR2	20	YES	Previous role
NEW_ROLE	VARCHAR2	20	YES	New role
PERFORMED_BY	NUMBER	22	NO	Admin user
PERFORMED_AT	TIMESTAMP	11	NO	Action time
IP_ADDRESS	VARCHAR2	45	YES	Client IP
USER_AGENT	VARCHAR2	500	YES	Browser info
NOTES	VARCHAR2	500	YES	Comments

OTP_CODES

Column	Type	Length	Nullable	Description
ID	NUMBER	22	NO	Primary Key
USER_ID	NUMBER	22	NO	FK → USERS
OTP_CODE	VARCHAR2	10	NO	Verification code
OTP_TYPE	VARCHAR2	20	NO	SMS/EMAIL/WHATSAPP
TARGET	VARCHAR2	100	NO	Phone/Email
EXPIRES_AT	TIMESTAMP	11	NO	Expiration time
IS_USED	NUMBER	22	YES	Used flag
CREATED_AT	TIMESTAMP	11	YES	Generation time

BLACKOUT_DAYS

Column	Type	Length	Nullable	Description
ID	NUMBER	22	NO	Primary Key
BLACKOUT_DATE	DATE	7	NO	Closure date
REASON	VARCHAR2	200	YES	Closure reason

Foreign Key Relationships

Parent Table	Child Table	Relationship
USERS	MEMBERSHIPS	User has memberships
USERS	USER_ROLE_MAP	User has roles
USERS	USER_GYM_ROLES	User has gym-specific roles
USERS	CHECK_INS	User makes check-ins
USERS	CLASS_BOOKINGS	User books classes
USERS	NOTIFICATIONS	User receives notifications
USERS	PT_SESSIONS	User has PT sessions
USERS	TRANSACTIONS	User makes transactions
USERS	GYM_STAFF	User is staff
USERS	TRAINER_CUSTOMER_MAP	Trainer-member pairing
USERS	ROLE_APPLICATIONS	User applies for roles
USERS	OTP_CODES	User has OTP codes
GYMS	MEMBERSHIPS	Gym has memberships
GYMS	CLASSES	Gym hosts classes
GYMS	GYM_STAFF	Gym employs staff
GYMS	GYM_SETTINGS	Gym has settings
GYMS	USER_GYM_ROLES	Gym has role assignments
GYMS	TRANSACTIONS	Gym tracks transactions
GYMS	ROLE_APPLICATIONS	Gym receives applications
ROLES	USER_ROLE_MAP	Role assigned to users
MEMBERSHIP_PACKAGES	MEMBERSHIPS	Package used in memberships
CLASSES	CLASS_BOOKINGS	Class has bookings

Key Statistics

Metric	Value
Total Users	170
Active Memberships	103
Membership Packages	5
User-Role Mappings	158
Trainer-Customer Pairs	19
PT Sessions	6
Transactions	51
System Roles	4
Gyms	1
Gym Settings	10

Security Features

- 1. **OAuth Integration** - Google & Facebook login support
- 2. **OTP Verification** - SMS/Email/WhatsApp codes
- 3. **Role-Based Access** - Granular permission system
- 4. **Audit Trail** - Role change tracking
- 5. **Account Locking** - Security lockout capability

Report generated from Oracle 23ai FREE database